
Laminar Non-Premixed (Diffusion) Flames

Outline *Moving from premixed combustion which dealt with systems where the fuel and oxidiser were premixed and chemical rates governed combustion, we now look into scenarios where the fuel and oxidiser are not premixed and this mixing becomes the rate limiting step for combustion. In most of these systems, the process of diffusion dominates the mixing phenomenon and therefore, non-premixed flames are also referred to as diffusion flames. To study this non-premixed behavior, we would be firstly looking into a non reacting jet as it diffuses in still air. Following the analysis of this non-reacting system, we look at the changes that comes with the inclusion of combustion reactions. Finally, some important empirical and theoretical results concerning laminar jet diffusion flames are discussed.*

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1 Introduction

In a *diffusion flame*, combustion occurs at the interface between the fuel gas and the oxidant gas, and the burning depends more on rate of diffusion of reactants than on the rates of chemical processes involved. This is because the rates of mixing are lower than the chemical reaction rates and therefore, becomes the bottleneck in this scenario. They are, therefore characterised by higher Damköhler numbers (defined as the ratio of diffusion or mixing times to reaction times). General examples include candle flames, a fuel droplet burning in air, a solid particle burning in oxygen rich environments etc. Such non-premixed flame can either be laminar or turbulent depending on the flow properties; in this chapter, we shall concentrate on laminar non-premixed flames. Compared to premixed flames, it is more difficult to give a general treatment of diffusion flames, largely because no simple, measurable parameter, analogous to the burning velocity in premixed flames, can be defined.

1.1 Phenomenological Analysis

For simple laminar diffusion flames on circular nozzles (similar to a candle flame), flame height is mostly used to characterize the flame. For simple treatments, the fundamental assumption that needs to be made here is that the reaction rates are controlled by the rates of diffusion and therefore, as soon as the reactants mix together, they react. For this radially axisymmetric case, there are two characteristic times that we have to consider; one is associated with the diffusion of oxidiser in the radial direction from surrounding to the centreline, the other is the convection time associated with the axial movement of fuel from the nozzle exit port to the tip of the flame. At the flame height, one can consider that the fuel is completely burned due to the diffusion of the required oxidiser from the surrounding region.

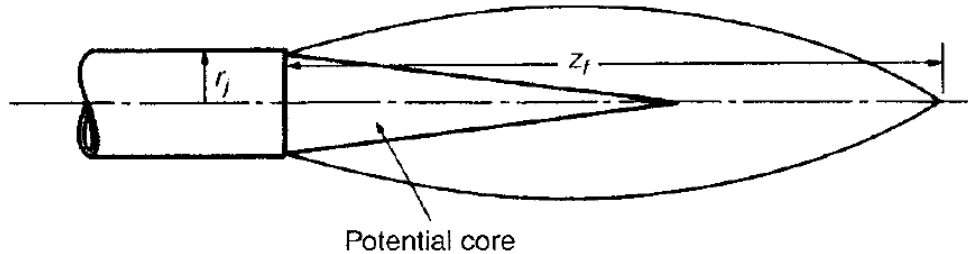


Figure 1: A horizontal laminar fuel jet

Now, based on the kinetic theory of gases, the average transverse displacement of a molecule due to molecular diffusion is given by :

$$\bar{y}^2 = 2Dt \quad (1)$$

where $\bar{y}^2$ denotes the mean squared displacement of a molecule from its initial location during time t . The length of the flame is assumed to correspond to the condition that at the point on the stream axis where fuel is completely consumed, the average depth of penetration of the air into fuel must

be approximately equal (or proportional, if one likes) to the radius of the burner tube. Therefore, approximating $\bar{y}^2$ by R^2 yields

$$t = R^2/2\mathcal{D} \quad (2)$$

The gas exit velocity is taken as a constant so that the flow residence time t_{res} from the nozzle exit to the tip of the flame is given by:

$$t_{res} = L_f/v \quad (3)$$

where v is the constant nozzle exit velocity. Now, assuming that the radial diffusion time for the oxidiser is of the same order of magnitude as the residence time of the fuel at flame height, we get

$$L_f \approx \frac{vR^2}{2\mathcal{D}} \approx \frac{\pi vR^2}{2\pi\mathcal{D}} \approx \frac{Q_F}{2\pi\mathcal{D}} \quad (4)$$

where Q_F is the volumetric flowrate of fuel. Although very crude, this approximation permits us to draw certain predictions:

- At a given flow rate, flame height is independent of the burner diameter.
- Since the diffusion coefficient $\mathcal{D}$ is inversely proportional to pressure, the height of the flame is independent of pressure at given mass flow rate.
- Flame height is proportional to volume flow rate of fuel.

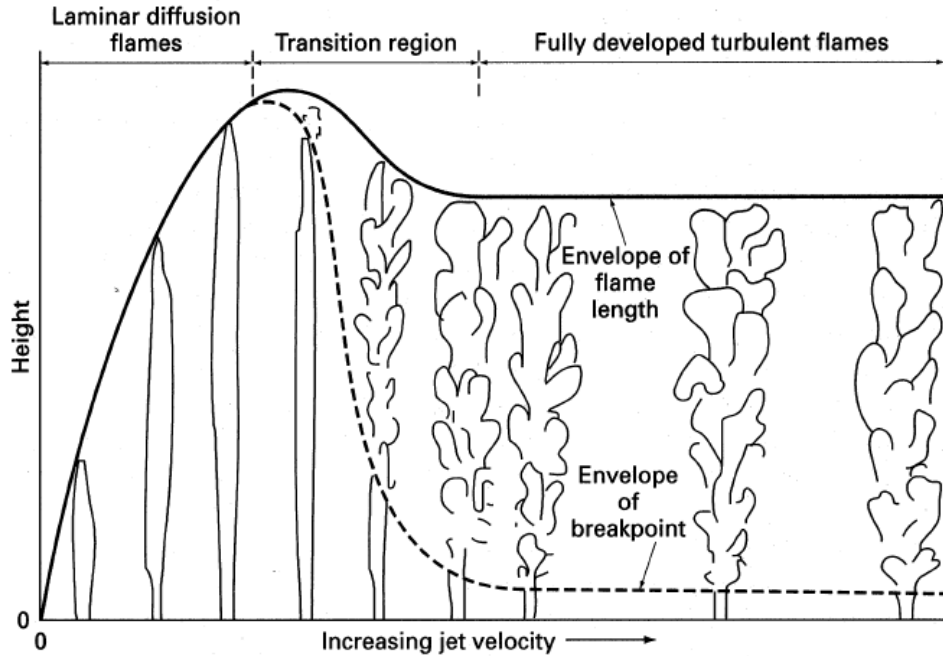


Figure 2: The variation of flame height and character as a function of jet velocity

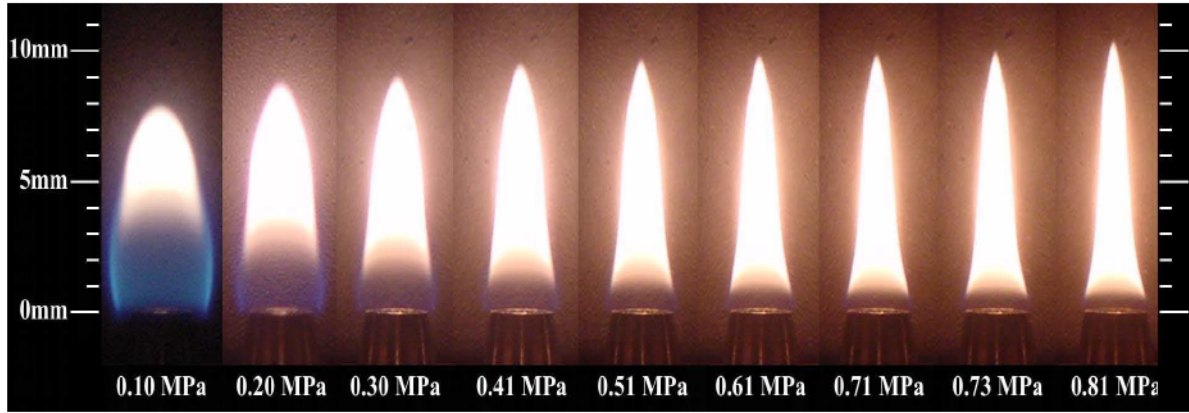


Figure 3: Propane diffusion flames at increasing pressure and same mass flowrate

1.2 Non-reacting Constant-Density Laminar Jet

Physical Description: The analysis presented in the previous section is very crude and only provides certain qualitative features of a laminar diffusion flame. Before moving to an analysis of reacting laminar jet, we start with a non-reacting laminar jet of a fluid flowing into an infinite reservoir. The important points to take away from this analysis that would still apply to the reacting case are the properties of the basic flow and diffusional processes.

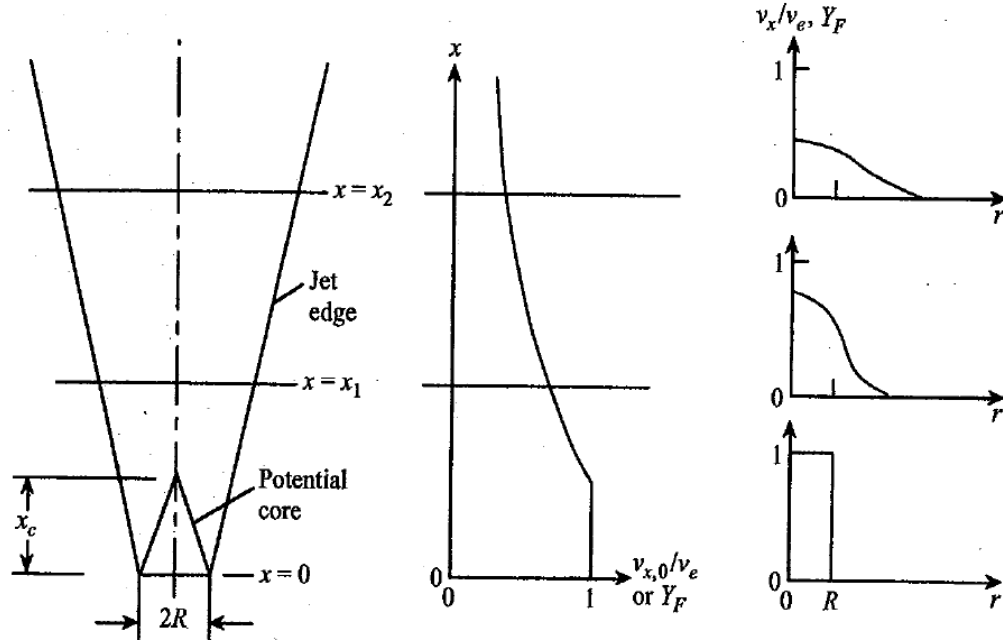


Figure 4: Non-reacting, laminar fuel jet issuing into an infinite reservoir of quiescent air

Figure 4 illustrates the essential features of a fuel jet issuing from a nozzle of radius R into still air. For simplicity, we assume that the velocity profile is uniform at the tube exit. Close to the nozzle, there exists a region called the potential core of the jet. In the potential core, the effects of viscous shear and diffusion have yet to be felt; hence, both the velocity and nozzle-fluid mass fraction remain unchanged from their nozzle-exit values and are uniform in this region. In the region between the potential core and the jet edge, both the velocity and fuel concentration decrease monotonically to zero at the jet edge. Beyond the potential core the viscous shear and diffusion effects are active across whole field of the jet.

Initial jet momentum is conserved throughout the entire flow-field. As the jet moves into surroundings, some of the momentum is transferred to air, decreasing the velocity of the jet, while greater and greater quantities of air are entrained into the jet as it proceeds downstream. We can express this mathematically using an integral form of momentum conservation:

$$\underbrace{2\pi \int_0^\infty \rho(r, x) v_x^2(r, x) r dr}_{\text{Momentum flow of the jet at any } x, J} = \underbrace{\rho_e v_e^2 \pi R^2}_{\text{Momentum flow issuing from the nozzle, } J_e} \quad (5)$$

where subscript e specifies the nozzle exit conditions. The process that control the diffusion and convection of momentum are similar to the processes that control the fuel concentration field (convection and diffusion of fuel mass). Consequently, distribution of fuel mass fraction, $Y_F(r, x)$, should be similar to dimensionless velocity distribution, $v_x(r, x)/v_e$. Fuel molecules diffuse radially outward according to Fick's law. As the fuel moves downstream, the time available for diffusion also increases and the width of the region containing fuel grows with x and the centerline fuel concentration decays. The mass of fluid issuing from nozzle is conserved throughout the flowfield:

$$2\pi \int_0^\infty \rho(r, x) v_x(r, x) Y_F(r, x) r dr = \rho_e v_e \pi R^2 Y_{F,e} \quad (6)$$

where $Y_{F,e} = 1$.

Assumptions: To determine the velocity and mass fraction fields we need to make some assumptions.

1. $MW_e = MW_\infty$. $P = \text{const.}$ $T = \text{const.}$: Uniform density field.
2. Species transport is by Fick's diffusion law.
3. Momentum and species diffusivities are constant and equal, i.e. the Schmidt Number is unity, $Sc \equiv \nu/\mathcal{D} = 1$
4. Diffusion is considered only in radial direction; axial diffusion is neglected. (This may not be a good assumption very near to the nozzle exit; since near the exit it is expected that the axial diffusion will be more significant in comparison with the downstream locations.)

Conservation Laws (Boundary layer equations)

- *Mass Conservation:*

$$\frac{\partial v_x}{\partial x} + \frac{1}{r} \frac{\partial(v_r r)}{\partial r} = 0 \quad (7)$$

- *Axial Momentum Conservation:*

$$v_x \frac{\partial v_x}{\partial x} + v_r \frac{\partial v_x}{\partial r} = \nu \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \quad (8)$$

- *Species Conservation:* For the jet fluid (fuel)

$$v_x \frac{\partial Y_F}{\partial x} + v_r \frac{\partial Y_F}{\partial r} = \mathcal{D} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_x}{\partial r} \right) \quad (9)$$

In addition, we have,

$$Y_{Ox} = 1 - Y_F \quad (10)$$

Boundary Conditions:

- To solve Eqns.(7), (8), (9) for the unknown functions $v_x(r, x)$, $v_r(r, x)$, and $Y_F(r, x)$ requires three boundary conditions each for v_x and Y_F , and one boundary condition for v_r .
- Along the jet centreline, $r = 0$,

$$v_r(0, x) = 0 \quad (11a)$$

$$\frac{\partial v_x}{\partial r}(0, x) = 0 \quad (11b)$$

$$\frac{\partial Y_F}{\partial r}(0, x) = 0 \quad (11c)$$

where the last two result from symmetry.

- At large radii ($r \rightarrow \infty$),

$$v_x(\infty, x) = 0 \quad (12a)$$

$$Y_F(\infty, x) = 0 \quad (12b)$$

- At the jet exit, $x = 0$, we assume uniform axial velocity and fuel mass fraction, and zero elsewhere:

$$\begin{aligned} v_x(r \leq R, 0) &= v_e \\ v_x(r > R, 0) &= 0 \end{aligned} \quad (13a)$$

$$\begin{aligned} Y_F(r \leq R, 0) &= Y_{F,e} = 1 \\ Y_F(r > R, 0) &= 0 \end{aligned} \quad (13b)$$

Solution: The velocity field can be obtained by assuming the profiles to be similar. The idea of similarity is that the intrinsic shape of the velocity profiles is the same everywhere in the flow-field. For the present problem, this implies that the radial distribution of $v_x(r, x)$, when normalized by the local centreline velocity $v_x(0, x)$, is a universal function that depends only on the similarity variable r/x . The solutions for axial and radial velocities are

$$v_x = \frac{3}{8\pi} \frac{J_e}{\mu x} \left[1 + \frac{\xi^2}{4} \right]^{-2} \quad (14)$$

$$v_r = \left(\frac{3J_e}{16\pi\rho_e} \right)^{1/2} \frac{1}{x} \frac{\xi - \frac{\xi^3}{4}}{\left(1 + \frac{\xi^2}{4} \right)^2} \quad (15)$$

where J_e is the jet initial momentum flow,

$$J_e = \rho_e v_e^2 \pi R^2 \quad (16)$$

and,

$$\xi = \left(\frac{3\rho_e J_e}{16\pi} \right)^{1/2} \frac{1}{\mu} \frac{r}{x} \quad (17)$$

Axial velocity distribution in dimensionless form (substitute Eqn.(16) into (??)),

$$\frac{v_x}{v_e} = 0.375 \left(\frac{\rho_e v_e R}{\mu} \right) \left(\frac{x}{R} \right)^{-1} \left[1 + \frac{\xi^2}{4} \right]^{-2} \quad (18)$$

Dimensionless centreline velocity decay obtained by setting $r = 0$ ($\xi = 0$),

$$\frac{v_{x,0}}{v_e} = 0.375 \left(\frac{\rho_e v_e R}{\mu} \right) \left(\frac{x}{R} \right)^{-1} \quad (19)$$

Velocity decays inversely with axial distance, and proportional to the jet Reynolds number,

$$\text{Re}_j \equiv \frac{\rho_e v_e R}{\mu}$$

From (19), we see that the solution is not valid near the nozzle; at small values of x , the dimensionless centerline velocity becomes larger than unity, which is not physically correct.

Other parameters frequently used to characterize jets are the spreading rate $r_{1/2}/x$ and spreading angle α . To define these parameters, we need to introduce the idea of the jet half-width, $r_{1/2}$. The jet half-width is simply the radial location where the jet velocity has decayed to one half of its centerline value. Jet spreading angle is the angle whose tangent is the spreading rate. An expression for $r_{1/2}$ can be derived by setting $v_x/v_{x,0}$ to be one half and solving for r .

$$r_{1/2}/x = 2.97 \left(\frac{\mu}{\rho_e v_e R} \right) = 2.97 \text{Re}_j^{-1} \quad (20)$$

$$\alpha \equiv \tan^{-1}(r_{1/2}/x) \quad (21)$$

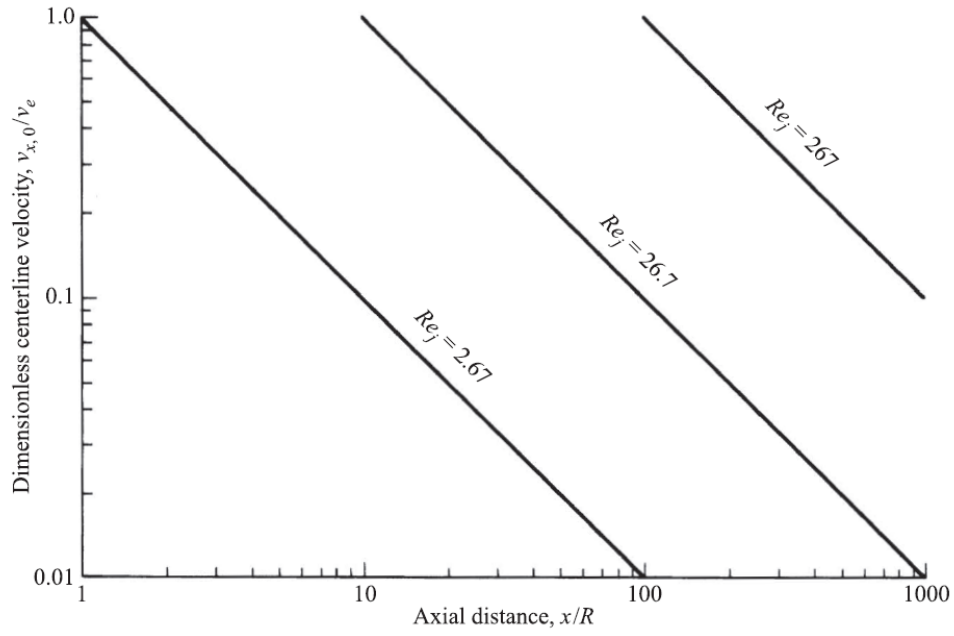


Figure 5: Centerline velocity decay for laminar jets.

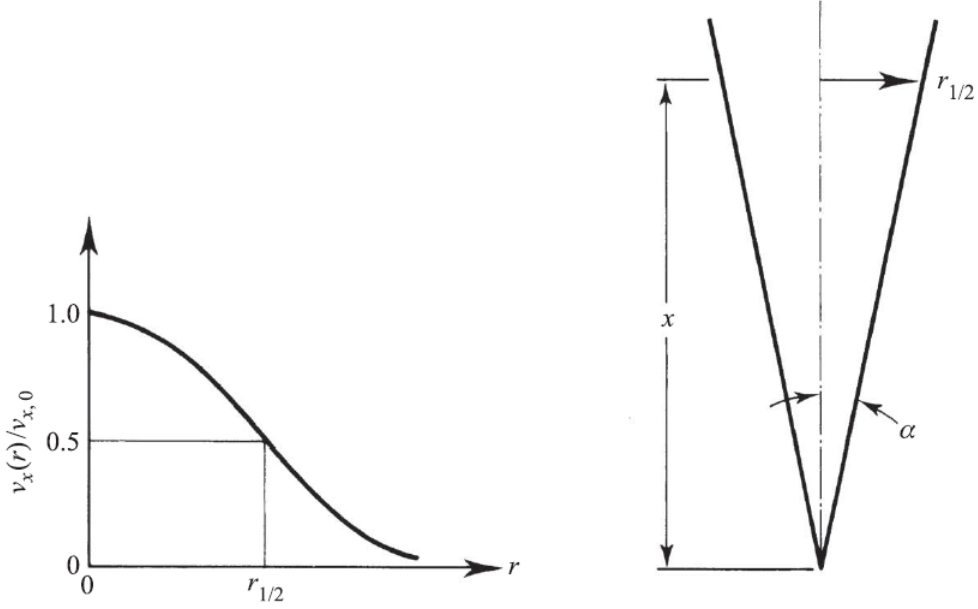


Figure 6: Definitions of jet half-width, $r_{1/2}$ and jet spreading angle, α

The equations (20), (21) reveals that high- Re_j jets are narrow, while low- Re_j jets are wide. Comparing Eqns. (8) and (9), we see that Y_F plays the same mathematical role as v_x/v_e , if the Schmidt

number is unity, i.e., $\nu = \mathcal{D}$. Then the functional form of the solution for Y_F is identical to that for v_x/v_e ,

$$Y_F = \frac{3}{8\pi} \frac{Q_F}{\mathcal{D}x} \left[1 + \frac{\xi^2}{4} \right]^{-2} \quad (22)$$

where $Q_F = v_e \pi R^2$, volumetric flow rate of fuel. By applying $Sc = 1$ to Eqn.(22)

$$Y_F = 0.375 \text{Re}_j \left(\frac{x}{R} \right)^{-1} \left[1 + \frac{\xi^2}{4} \right]^{-2} \quad (23)$$

Centreline values of mass fraction,

$$Y_{F,0} = 0.375 \text{Re}_j \left(\frac{x}{R} \right)^{-1} \quad (24)$$

Again, it should be noted that the solutions are valid far from the nozzle. The dimensionless distance downstream where the solution is valid must exceed the jet Reynolds number, that is,

$$(x/R) \geq 0.375 \text{Re}_j \quad (25)$$

1.3 Jet Flame Physical Description

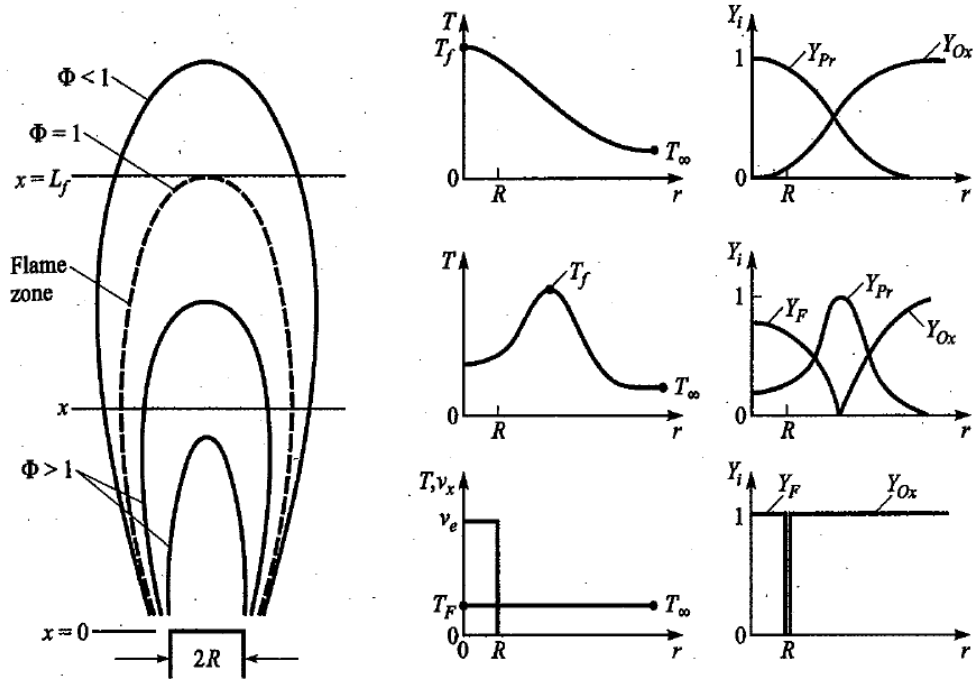


Figure 7: Laminar diffusion flame structure.

The burning laminar fuel jet has much in common with our previous discussion of the non-reacting jet. As the fuel flows along the flame axis, it diffuses radially outward, while the oxidizer diffuses

radially inward. The “flame surface” can be defined as the locus of all points where Φ equals unity. The products formed at the flame surface diffuse radially both inward and outward. An *over-ventilated flame* is where there is more than enough oxidizer in the immediate surroundings to continuously burn the fuel. Flame length for an over-ventilated flame is determined at the axial location where,

$$\Phi(r = 0, x = L_f) = 1 \quad (26)$$

The region where chemical reactions occur is generally quite narrow. As seen in Fig. 7, the high-temperature reaction zone occurs in an annular region until the flame tip is reached. In the upper regions of a vertical flame, there is a sufficient quantity of hot gases that buoyant forces become important. Buoyancy accelerates the flow and causes a narrowing of the flame, since conservation of mass requires streamlines to come closer together as the velocity increases. The narrowing of the flow increases the fuel concentration gradients, dY_F/dr , thus enhancing diffusion. The effects of these two phenomena on the length of flames issuing from circular nozzles tend to cancel. Thus, simple theories that neglect buoyancy fortuitously are able to predict flame lengths reasonably well for the circular- (and square-) port geometry.

By ignoring the effects of heat released by reaction, (22) provides a crude description of flame boundaries when $Y_F = Y_{F,stoic}$. When r equals zero, we get a flame length,

$$L_f \approx \frac{3}{8\pi} \frac{Q_F}{\mathcal{D}Y_{F,stoic}} \quad (27)$$

Thus, we see that the flame length is indeed proportional to the volumetric flowrate; furthermore, we see that the flame length is inversely proportional to the stoichiometric fuel mass fraction. This implies that fuels that require less air for complete combustion produce shorter flames, as one might expect. Since $Q_F = v_e \pi R^2$, various combinations of v_e and R can yield the same flame length. Also, since the diffusion coefficient $\mathcal{D}$ is inversely proportional to pressure, the height of the flame is independent of pressure at given mass flow rate.

2 Theoretical and Empirical Formulations

Owing to its great applicability in understanding flame chemistry and structure effects, coflow flames have been analysed by a number of researchers who have formulated theoretical and empirical formulations for quantifying the various properties of coflow diffusion flames. Some of the most historically important formulations are:

- Burke and Schumann (1928)
 - constant velocity field parallel to flame axis.
 - reasonable predictions of L_f for round burners.
- Roper and Roper et al (1977)
 - relaxed single constant velocity assumption.
 - provides extremely good predictions.
 - matched by experimental results/correlations.

– round and slot-burners.

Owing to the immense success of Roper’s relations for quantifying coflow flame properties, we will be using the same in our analysis.

2.1 Roper’s Solutions and Correlations:

Circular Port:

$$L_{f,\text{thy}} = \frac{Q_F(T_\infty/T_F)}{4\pi\mathcal{D}_\infty \ln(1 + 1/S)} \left(\frac{T_\infty}{T_f} \right)^{0.67} \quad (28)$$

$$L_{f,\text{expt}} = 1330 \frac{Q_F(T_\infty/T_F)}{\ln(1 + 1/S)} \quad (29)$$

where S is stoichiometric molar oxidizer-fuel ratio, $\mathcal{D}_\infty$ mean diffusion coefficient of oxidizer at T_∞ , T_F and T_f are fuel stream and mean flame temperatures, respectively.

Square Port:

$$L_{f,\text{thy}} = \frac{Q_F(T_\infty/T_F)}{16\mathcal{D}_\infty \{\text{inverf}[(1 + S)^{-0.5}]\}^2} \left(\frac{T_\infty}{T_f} \right)^{0.67} \quad (30)$$

$$L_{f,\text{expt}} = 1045 \frac{Q_F(T_\infty/T_F)}{\{\text{inverf}[(1 + S)^{-0.5}]\}^2} \quad (31)$$

where inverf is the inverse of error function Erf,

$$\text{Erf}w = \frac{2}{\sqrt{\pi}} \int_0^w e^{-t^2} dt \quad (32)$$

Slot Burner–Momentum Controlled:

$$L_{f,\text{thy}} = \frac{b\beta^2 Q_F}{hI\mathcal{D}_\infty Y_{F,\text{stoic}}} \left(\frac{T_\infty}{T_F} \right)^2 \left(\frac{T_f}{T_\infty} \right)^{0.33} \quad (33)$$

$$L_{f,\text{expt}} = 8.6 \cdot 10^4 \frac{b\beta^2 Q_F}{hIY_{F,\text{stoic}}} \left(\frac{T_\infty}{T_F} \right)^2 \quad (34)$$

where b is the slot width and h is the length, and,

$$\beta = \frac{1}{4 \times \text{inverf}[1/(1 + S)]} \quad (35)$$

I is the ratio of actual initial momentum flow from the slot to that of uniform flow,

$$I = \frac{J_{e,\text{act}}}{\dot{m}_F v_e} \quad (36)$$

For uniform flow $I = 1$. For a fully developed flow, assuming parabolic exit velocity, $I = 1.5$. Equations and are only applicable to conditions where the oxidizer is stagnant.

Slot Burner–Buoyancy Controlled:

$$L_{f,\text{thy}} = \left[\frac{9\beta^4 Q_F^4 T_\infty^4}{8\mathcal{D}_\infty^2 a h^4 T_F^4} \right]^{1/3} \left(\frac{T_f}{T_\infty} \right)^{2/9} \quad (37)$$

$$L_{f,\text{expt}} = 2 \cdot 10^3 \left[\frac{\beta^4 Q_F^4 T_\infty^4}{a h^4 T_F^4} \right]^{1/3} \quad (38)$$

where a is the mean buoyant acceleration,

$$a \cong 0.6g \left(\frac{T_f}{T_\infty} - 1 \right) \quad (39)$$

and g is the gravitational acceleration.

Slot Burner–Transition Regime: *Froude Number*,

$$\text{Fr}_f \equiv \frac{(v_e I Y_{F,\text{stoic}})^2}{a L_f} \quad (40)$$

Froude number physically represent the ratio of the initial jet momentum flow to the buoyant force experienced by the flame.

- $\text{Fr}_f \gg 1$: Momentum-controlled
- $\text{Fr}_f \approx 1$: Transition (mixed)
- $\text{Fr}_f \ll 1$: Buoyancy-controlled

Note that L_f must be known a priori to establish the appropriate regime. So it requires a trial and error approach. When $\text{Fr}_f \approx 1$,

$$L_{f,T} = \frac{4}{9} L_{f,M} \left(\frac{L_{f,B}}{L_{f,M}} \right)^3 \left\{ \left[1 + 3.38 \left(\frac{L_{f,M}}{L_{f,B}} \right)^3 \right]^{2/3} - 1 \right\} \quad (41)$$